

School of Engineering and Applied Science (SEAS), Ahmedabad University

CSE 400: Fundamentals of Probability in Computing
Lecture-19 Scribe Submission

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CSE400 – Lecture 19 Scribe

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1. Joint Gaussian Distribution

Definition:

$$f_{XY}(x, y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \exp\left(-\frac{1}{2(1-\rho^2)}\left[\left(\frac{x-\mu_X}{\sigma_X}\right)^2 + \left(\frac{y-\mu_Y}{\sigma_Y}\right)^2 - 2\rho\left(\frac{x-\mu_X}{\sigma_X}\right)\left(\frac{y-\mu_Y}{\sigma_Y}\right)\right]\right)$$

Notation:

- μ_X, μ_Y : means
- σ_X, σ_Y : standard deviations
- ρ : correlation coefficient

2. Example: Joint Gaussian

Given:

$$\mu_X = 2, \quad \sigma_X^2 = 1 \Rightarrow \sigma_X = 1$$

$$\mu_Y = -3, \quad \sigma_Y^2 = 4 \Rightarrow \sigma_Y = 2$$

$$\text{cov}(X, Y) = -1$$

Step 1: Correlation

$$\rho = \frac{\text{cov}(X, Y)}{\sigma_X\sigma_Y} = \frac{-1}{1 \cdot 2} = -\frac{1}{2}$$

Step 2: Substitution Substitute into the joint Gaussian formula to obtain the joint PDF.

3. Marginal PDFs

Definition:

$$f_X(x) = \int_{-\infty}^{\infty} f_{XY}(x, y) dy, \quad f_Y(y) = \int_{-\infty}^{\infty} f_{XY}(x, y) dx$$

Result:

$$X \sim \mathcal{N}(2, 1), \quad Y \sim \mathcal{N}(-3, 4)$$

$$f_X(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{(x-2)^2}{2}}$$

$$f_Y(y) = \frac{1}{2\sqrt{2\pi}} e^{-\frac{(y+3)^2}{8}}$$

4. Probability Computation

Step 1:

$$P(X < 0) = F_X(0)$$

Step 2: Standardization

$$Z = \frac{X - 2}{1} \Rightarrow P(X < 0) = P(Z < -2)$$

Step 3: Symmetry

$$P(Z < -2) = P(Z > 2)$$

Final Result:

$$P(X < 0) = Q(2)$$

5. Joint Distribution of N Variables

$$\mathbf{X} = \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_N \end{bmatrix}$$

Joint distribution is described using:

- Joint PMF
- Joint PDF
- Joint CDF

6. Joint Functions

Joint PMF:

$$P(X_1 = x_1, \dots, X_N = x_N)$$

Joint CDF:

$$F(x_1, \dots, x_N) = P(X_1 \leq x_1, \dots, X_N \leq x_N)$$

Joint PDF:

$$f(x_1, \dots, x_N) = \frac{\partial^N F}{\partial x_1 \cdots \partial x_N}$$

7. Independence of Random Variables

$$f_{X_1, \dots, X_N}(x_1, \dots, x_N) = \prod_{i=1}^N f_{X_i}(x_i)$$

8. Example: Uniform Distribution (3D)

Given:

$$f(x_1, x_2, x_3) = \begin{cases} C, & |x_i| \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

Step 1: Find C

$$\int_{-1}^1 \int_{-1}^1 \int_{-1}^1 C \, dx_1 dx_2 dx_3 = 1$$

$$C \cdot 2 \cdot 2 \cdot 2 = 1 \Rightarrow C = \frac{1}{8}$$

Step 2: Marginals

$$f_{X_1, X_2}(x_1, x_2) = \frac{1}{4}, \quad f_{X_1}(x_1) = \frac{1}{2}$$

Step 3: Independence

$$f_{X_1, X_2, X_3} = \frac{1}{8}$$

$$f_{X_1} f_{X_2} f_{X_3} = \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{8}$$

Conclusion: Independent

9. Example: Trivariate Exponential

Given:

$$f_{XYZ}(x, y, z) = \begin{cases} K e^{-(ax+by+cz)}, & x, y, z > 0 \\ 0, & \text{otherwise} \end{cases}$$

Step 1: Find K

$$K \int_0^\infty e^{-ax} dx \int_0^\infty e^{-by} dy \int_0^\infty e^{-cz} dz = 1$$

$$K \cdot \frac{1}{a} \cdot \frac{1}{b} \cdot \frac{1}{c} = 1 \Rightarrow K = abc$$

Step 2: Marginals

$$f_X(x) = a e^{-ax}$$

Step 3: Independence

$$f_{XYZ} = abce^{-(ax+by+cz)}$$

$$f_X f_Y f_Z = abce^{-(ax+by+cz)}$$

Conclusion: Independent